Mat2002 Tutorial 4

Junwei Zha

February 2, 2026

Second Order linear ODE

- General form of second order linear ODE is :

$$y'' + p(t)y' + q(t)y = g(t)$$

- Denote L to be the operator with $L(y) = y'' + p(t)y' + q(t)y$.
- The ODE is called Homogeneous if $g(t) \equiv 0$. The equation $L(y) = 0$ is called the associated homogeneous equation to $L(y) = g(t)$.
- Initial condition: $y(t_0) = y_0, y'(t_0) = y_0'$.

Existence and Uniqueness Theorem

Theorem (Existence and Uniqueness)

Consider IVP:

$$\begin{cases} y'' + p(t)y' + q(t)y = g(t) \\ y(t_0) = y_0, y'(t_0) = y_0'. \end{cases}$$

If $p(t), q(t), g(t)$ are continuous in an open interval I containing t_0, then there exists unique solution in I.

Remark

The proof relies on existence of uniqueness for first order linear ODE system.

Existence and Uniqueness Theorem

Example

$$\begin{cases} \left(t^2-3t\right)y''+y'-(t+3)y=e^t \\ y(1)=2, \quad y'(1)=1. \end{cases}$$

Sol: Rewrite $y''+\frac{1}{t^2-3t}y'-\frac{t+3}{t^2-3t}y=\frac{e^t}{t^2-3t}, \quad t\neq 0,3.$
Since $t_0=1$, the solution uniquely exists in $(0,3)$.

Example

$$\begin{cases} y''+\sin\left(t^2\right)y'+\ln\left(e^t+2\right)y=0 \\ y(100)=0, \quad y'(100)=0 \end{cases}$$

has a unique solution $y=0$ on $\mathbb{R}$

General Theory for Homogeneous Equation & Wronskian

We will study the structure of solution to homogeneous equation:

$$L(y) = y'' + p(t)y' + q(t)y = 0$$

Theorem (Principle of superposition, or Linearity)

If $y_1(t), y_2(t)$ are solutions to $y'' + p(t)y' + q(t)y = 0$, then the linear combination $y = c_1y_1(t) + c_2y_2(t)$ are also solutions, $c_1, c_2 \in \mathbb{R}$.

Remark

This theorem means that the solution to the above ODE is a vector space

Question

What is the dimension of that vector space? What is the basis?

Linearly independent

Definition

Two functions $f(t), g(t), t \in I$ are linearly independent if $k_1 f_1(t) + k_2 f_2(t) = 0, \forall t \in I$ only hold for $k_1 = k_2 = 0$.

E.g. $\begin{cases} f_1(t) = t, f_2(t) = t^2 & t \in [0,1] \text{ are linearly independent} \\ f_1(t) = t, f_2(t) = 2t & t \in [0,1] \text{ are linearly dependent.} \end{cases}$

Definition (Wronskian)

Given two functions $f(t), g(t)$, the Wronskian of f, g is

$$W(f,g)(t) = \begin{vmatrix} f(t) & g(t) \\ f'(t) & g'(t) \end{vmatrix} = f(t)g'(t) - f'(t)g(t)$$

Linearly independent

Lemma

Let $f(t), g(t)$ be defined on I, if $W(f,g)(t_0) \neq 0$ for some $t_0 \in I$, then $f(t), g(t)$ are linearly independent.

Remark

The converse is not true for general functions. i.e.
f, g are linearly independent $\nRightarrow W(f,g)(t_0) \neq 0$ for some t_0.
e.g. $f(t) = t^3, g(t) = |t|^3$ are linearly independent, but $W(f,g)(t) \equiv 0$.

Theorem

Let $y_1(t), y_2(t)$ be two solutions to

$$y'' + p(t)y' + q(t)y = 0, \quad t \in I.$$

where $p(t), q(t)$ are continuous on I. Then y_1, y_2 are linearly independent on $I \Leftrightarrow W(y_1, y_2)(t_0) \neq 0$ for some $t_0 \in I$.

Structure of solution set for homogeneous equation

Theorem

Consider $y'' + p(t)y' + q(t)y = 0$, *where* $p(t), q(t)$ *are continuous on* I. *Let* $y_1(t), y_2(t)$ *be any two solutions with* $W(y_1, y_2)(t_0) \neq 0$, *for some* $t_0 \in I$.

Then:

- *The general solutions are* $y(t) = c_1 y_1(t) + c_2 y_2(t), c_1, c_2 \in \mathbb{R}$. $y_1(t), y_2(t)$ *are called a fundamental set of solutions to the homogeneous ODE.*
- *If the initial condition is prescribed at* $t_0 : y(t_0) = y_0, y'(t_0) = y_0'$, *then* c_1, c_2 *can be uniquely determined.*

Abel's theorem

Theorem

If $y_1(t), y_2(t)$ are two solutions of $y'' + p(t)y' + q(t)y = 0$, where $p(t), q(t)$ are continuous in I, then

$$W(y_1, y_2)(t) = ce^{-\int p(t)dt} \quad \text{for some } c \in \mathbb{R}.$$

Remark

As a consequence, Wronskian of two solutions y_1, y_2 to the homogeneous equation is either always zero ($c = 0$), or never zero ($c \neq 0$).

Application of Abel's theorem

Example

Consider $2t^2y'' + 3ty' - y = 0, t > 0$. One solution $y_1 = t^{\frac{1}{2}}$ is given, find another solution $y_2(t)$ s.t. y_1, y_2 form the fundamental set of solution.

Sol: $y'' + \frac{3}{2t}y' - \frac{1}{2t^2}y = 0, \quad t > 0$ By Abel's theorem,

$$W(y_1, y_2)(t) = \begin{vmatrix} t^{\frac{1}{2}} & y_2 \\ \frac{1}{2}t^{-\frac{1}{2}} & y_2' \end{vmatrix} = ce^{-\int \frac{3}{2t}dt} = ct^{-\frac{3}{2}}.$$

Let's choose $c = 1$ (any nonzero number is OK!)

$$t^{\frac{1}{2}}y_2' - \frac{1}{2}t^{-\frac{1}{2}}y_2 = t^{-\frac{3}{2}}, \quad t > 0.$$

It is a first order linear ODE. Let $\mu(t) = e^{-\int \frac{1}{2t}dt} = t^{-\frac{1}{2}}$. then

$y_2 = t^{\frac{1}{2}}\left(\int t^{-\frac{5}{2}}dt + c\right) = -\frac{2}{3}t^{-1}$ (choose $c = 0$ for simplicity)

Excercise I

Compute the Wronskian of the given pair of functions

(1) $e^t \sin t, e^t \cos t$

(2) $\cos^2 \theta, 1 + \cos 2\theta$

Excercise I

Compute the Wronskian of the given pair of functions
(1) $e^t \sin t, e^t \cos t$
(2) $\cos^2 \theta, 1 + \cos 2\theta$

Answer:(1) $-e^{2t}$ (2) 0

Exercise II

If y_1,y_2 are two solutions of

$$ty'' + 2y' + te^t y = 0, t > 0$$

and

$$W(y_1, y_2)(1) = 2$$

Find $W(y_1, y_2)(5)$

Exercise II

If y_1,y_2 are two solutions of

$$ty'' + 2y' + te^t y = 0, t > 0$$

and

$$W(y_1, y_2)(1) = 2$$

Find $W(y_1, y_2)(5)$

Sol: Rewrite $y'' + 2/ty' + e^t y = 0$
By Abel's theorem: $W(y_1, y_2)(t) = ce^{\int -\frac{2}{t} dt} = ct^{-2}$ $W(y_1, y_2)(1) = 2$ implies $c = 2$
Hence $W(y_1, y_2)(5) = \frac{2}{25}$